um-game-playing-strength-of-mcts-variants

3rd place solution - two stage flip augmentation stacking with code

Rank	3
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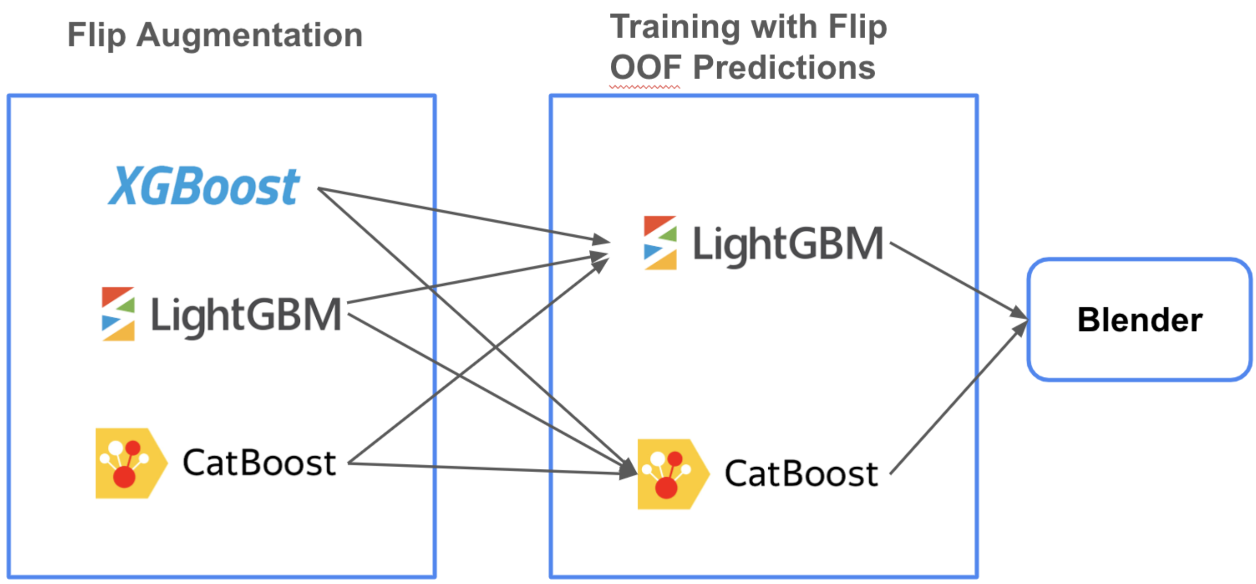
Version	Original

Thanks to UM organizers and kaggle team giving us such a interesting competition.This competition has no leak,stable cv-lb correlation,and lots of possibilities to improve, I really enjoyed exploring the accuracy boundaries of infinite possibilities.

Overview

I think the key point is data augmentation and creating a stable cv lb pipeline(lb is more important), the private leaderboard has almost no shake show us public and private dataset should be the same distribution.



Data Augmentation

filp the agent1-agent2 pair to agent2-agent1,change the AdvantageP1 to 1-AdvantageP1,change utility_agent1 to -utility_agent1,others keep same.

Cross Validation

StratifiedGroupKFold by GameRulesetName, modify minor class of utility_agent1 to neighbour class.

Feature Engingeering

I add some features from this great notebook https://www.kaggle.com/code/yunsuxiaozi/mcts-starter
And I also add tfidf-svd features of EnglishRules,this feature need to find the better parameter of max_features,I tried many times by lb score.

Feature Selection

I like use null importance feature selection to drop those features will change too much if target shift

Hyper-parameter tuning

I use optuna to search best hyper-parameters for lightgbm and catboost

Models

I use two stage modeling,the first stage is using flip augmentation data as train data to predict original train data and test data,then generate oof predictions as the second stage's feature.

Ensemble

My final submission is blending lightgbm and catboost with 3 seeds average ensemble.

Post processing

The final prediction mean values are lower than original mean values, same with public lb, so I multiply coefficients(1.12) can improve lb score of 0.002.

Not work

LudRules tfidf, bert embeddings,pseudo label with more agent1-agent2-games groups,neural network

Notebook

https://www.kaggle.com/code/senkin13/um-final-sub-1

Supplemental Q&A

> **yunsuxiaozi · 2024-12-03T02:30:10.827000**
>
> Congratulations, based on your experience participating in the competition, would it be better to add oof_pred as a feature to the data?

> **senkin13 · 2024-12-03T02:44:36.220000**
>
> yes,better than just add augmentation to train

> **Athar Sayed · 2024-12-03T04:24:26.607000**
>
> Congragulations . It was amazing to see the two stage flip augmentation from LMSYS competition applied here as well 😀. Also if you can share how did you find the coefficient 1.12 to multiply

> **senkin13 · 2024-12-03T07:48:32.547000**
>
> leaderboard probing, try different coefficients to see which has best lb score

> **Cafelatte1 · 2024-12-03T07:52:02.170000**
>
> Great solution :)
> Do the models in stage1 predict all features? or reduced dimension with SVD?

> **senkin13 · 2024-12-03T10:35:09.803000**
>
> stage1 predict the same target with stage2(utility_agent1)

> **mavillan · 2024-12-03T22:32:44.960000**
>
> Hi @senkin13 another related question: did you use any tricks to make stacking work? Like nested cross-validation or something?

> **senkin13 · 2024-12-04T03:18:53.473000**
>
> just use StratifiedGroupKFold by GameRulesetName at stage1 too